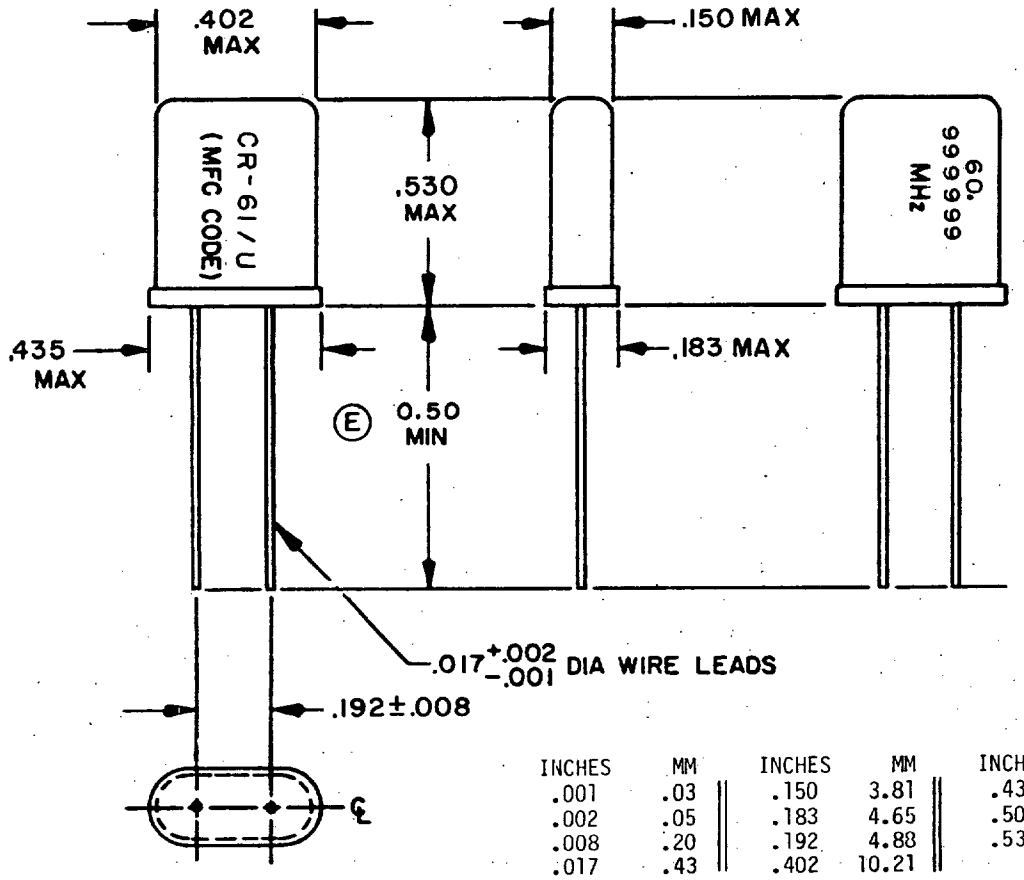
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-61/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 17 to 61 MHz; third overtone; controlled; series.



INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.150	3.81	.435	11.05
.002	.05	.183	4.65	.50	12.7
.008	.20	.192	4.88	.530	13.46
.017	.43	.402	10.21		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type-designation, frequency, and code-designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Dimensions and configuration.

Ⓔ denotes changes

FSC 5955

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TABLE I. Requirements.

Requirements	Specified value or condition
(E) Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure
Frequency range	17 to 61 MHz, inclusive
Capacitance, shunt	7 pF, maximum
Frequency tolerances:	
Operating temperature range	±20 PPM
Room temperature	±80 PPM
Frequency stability	±5 PPM
Equivalent resistance	40 ohms, maximum
Mode of oscillation	Third mechanical overtone
Reference temperature	85°C ±1°C
Temperature ranges:	
Operating	80°C to 90°C, inclusive
Operable	-55°C to +80°C, inclusive
Test set, calibration values, and rated drive level	See table II
Shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 PPM

1/ Method 201 of MIL-STD-202.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies inclusive	Calibration values		Rated drive level
		Resistance	Resistor voltage drop	
	MHz	Ohms	Volts	mW
TS-683/TSM	17 to 25	40	0.28	2.0 ±0.4
	25+ to 61	40	0.20	1.0 ±0.2

NOTE: NATO preferred type NXT22.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

Users activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0623-5)

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